

Subject Description Form

Subject Code	AF3214
Subject Title	Python Programming for Accounting and Finance
Credit Value	3
Level	3
Normal Duration	One Semester
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: Financial Accounting (AF2108) and Business Finance (AF3313)
Objectives	This subject is designed to study the scientific computing skills and apply the skills to solve accounting and finance problems faced in real-life situations. The subject introduces the application of Python in accounting and finance and uses Python to value different financial products using API and other methods. This subject also covers basic modeling, basic text mining and web scraping using Python. After studying this subject, students should master the necessary computing tools for further study and work in the fields of accounting and finance.
Intended Learning Outcomes (Note 1)	Upon successful completion of this subject, students should be able to: a. Confidently tackle basic coding tasks and build a foundation for further learning in accounting and finance analytics b. Master the basic scientific computing skills and acquire basic knowledge of web scraping, understand relational database systems c. Understand the risk measures and their calculation using real data d. Achieve direct market access by utilizing the Application Programming Interface (API) e. Analyze financial disclosures for financial risk assessment using textual analysis or large language model f. Understand data ethics by systemizing, defending, and recommending concepts of right and wrong conduct in relation to personal data.
Subject Synopsis/ Indicative Syllabus (Note 2)	Python Bootcamp Covers a range of fundamental programming concepts, including variables and data types, using control flow mechanisms like loops and conditional statements, defining and calling functions, and lists.

	Libraries like Pandas for data manipulation and analysis can equip students with practical tools for various programming tasks. Introduction to Database Systems Database systems fundamentals, including relational model, relational algebra operations, alternative data models, vector databases, as well as modern SQL features and common table expressions. Basic Scientific Computing Issues in the use of scientific computing software packages in Python, including text mining, web scrapping, and downloading stock data via Application Programming Interface (API). Direct Market Access (DMA) Utilize the Application Programming Interface (API) to achieve direct market access, and understand the basic order types. Risk Measures Understand the risk measures and their calculation using scientific computing based on real market data. Textual Analysis & Large Language Model in Accounting and Finance Explore the application of textual analysis techniques and large language models to analyze financial disclosures so as to assess financial risks and gain a deeper understanding of market sentiment, ultimately informing accounting decisions and financial reporting. Ethical Issues Ethical issues in applying technological tools to business.
Teaching/Learning Methodology (Note 3)	Key concepts and techniques will be introduced through lectures and a lot of emphases will be placed on individual assignment or project work. Students will be required to deliver an individual assignment or a project with a focus on real-world accounting and finance issues. By completing the work, students should have hands-on experience in using the knowledge they have learnt in class to solve accounting and finance problems in practice. Students are also encouraged to share their views and experiences actively with their lecturer and classmates.

Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					
			a	b	c	d	e	f
	1. Quizzes	10%	√	√	√	√	√	√
	2. Individual Assignments	40%	√	√	√	√	√	√
	3. Midterm	20%	√	√				
	4. Final Examination	30%	√	√	√	√	√	
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Quizzes and Individual Assignment – are used to test students’ understanding of the teaching materials and achievement of the intended learning outcomes through a more in-depth investigation of different topics. Midterm – Midterm is used to test students’ skills and knowledge what has been taught in class. Final Examination – Final examination is used to test students’ overall ability in applying the knowledge learnt in the subject. <i>Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.</i>								
Student Study Effort Expected	Class contact:							
	▪ Seminars						39 Hours	
	Other student study effort:							
	▪ Self-study including reading textbook and preparation for quizzes, and examination						55 Hours	
	▪ Preparation for the individual assignment or project						20 Hours	
	Total student study effort						114 Hours	

Reading List and References	<i>Algorithmic Trading and DMA: An introduction to direct access trading strategies</i>, Latest Edition, Barry Johnson, 4Myeloma Press. <i>Python for Finance: Apply powerful finance models and quantitative analysis with Python</i>, 2nd Edition, by Yuxing Yan <i>How to Think Like a Computer Scientist - Learning with Python</i>, by Allen Downey, Jeff Elkner and Chris Meyers. Green Tea Press (https://greenteapress.com/wp/learning-with-python/) Some additional readings will be distributed in class.
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Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.

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